ADVIKA NAGOOL

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SUMMARY

Data Science student exploring Machine Learning, AI, and Quantum Computing, with experience in developing AI systems, applied research, and data-driven pipeline solutions. Co-author of a Springer-submitted research paper, with projects spanning intelligent systems and machine learning. National gold medallist in archery, bringing precision, discipline, and a structured problem-solving approach to technology.

EDUCATION

Mukesh Patel School of Technology Management & Engineering, Mumbai *2024 – 2029*
MBA + BTech, Data Science — CGPA: 9.4 / 10

Pace Junior Science College, Mumbai *2022 – 2024*
Class XII: 88.50%

I.E.S Orion, Mumbai *2010 – 2022*
Class X: 95.40%

TECHNICAL SKILLS

Languages: Python, C++, C-family (Java, Kotlin), R, JavaScript, TypeScript

Scientific Computing & Machine Learning: NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, PyTorch, TensorFlow, OpenCV

Data & Analysis: MySQL, MongoDB, Excel, Power BI, Tableau

Backend & Interfaces: FastAPI, Node.js, React, LangChain, LangGraph, Firebase

Prototyping & Tools: Git/GitHub, VS Code, Android Studio, AutoCAD, Jupyter

EXPERIENCE

Marketing Makhni | Research & Development Intern *Mar 2026 – Aug 2026*

- Conducted competitive analysis for 10+ companies, gathering and structuring market, competitor, and pricing data through web research, scraping, and client-side data acquisition to support business and strategic decision-making.
- Led client-side research projects from requirement gathering to final delivery, coordinating with stakeholders, managing data acquisition workflows, and ensuring timely delivery of analysis-ready datasets and project outputs.

Adihuman AI | AI Research Intern *Dec 2025 – Mar 2026*

- Developed a structured evaluation rubric for GAIA Level 3 tasks, covering multiple failure modes and enabling consistent assessment of complex AI agent outputs through clearly defined evaluation criteria.
- Evaluated multi-step LangGraph conversational workflows across 50+ edge cases, improving consistency in tone, voice, and emotional register through prompt frameworks with explicit constraints.

PROJECTS

Multi-Agent Market Analysis System — Evidence-Gated Brand Recommendations *2026*

- Built a four-agent Python system (Research → Analysis → Investigation → Control) that answers brand-attention questions from live APIs (Wikipedia, Media Cloud, YouTube, Wayback), detecting change points with Welch/Poisson tests under FDR correction and clustering them into cross-source events.
- Added a deterministic citation gate that rejects any LLM claim without a matching retrieved record, so the system returns "insufficient evidence" instead of inventing an explanation; verified with 600+ tests.

Stoch Claim — Probabilistic Sequence Modelling & Decision Pipeline *2026*

- Built a decision pipeline that routes a claim to the right action in under 7 seconds, using Hidden Markov Models for latent-state sequence estimation and Markov Decision Processes for value-iteration routing over 1,000 records
- Combined stochastic modelling (HMM, MDP, Gambler's Ruin) with LLM-generated explanations (Llama-3.3-70B via Groq) for cost-optimal routing across five action classes with full audit trails

Braille Bridge — Noise-Robust Tactile Signal Processing & Recognition *2025*

- Built a word-level tactile Braille recognition pipeline, deployed via Gradio and React, that reads raw sensor spike trains from the iCub taxel-array dataset and predicts character sequences without frame-level alignment
- Cleaned raw sensor input with a 1D convolutional denoising autoencoder ahead of a CNN-LSTM with CTC decoding, and isolated the denoising stage's contribution through ablation studies

RESEARCH PAPER & PUBLICATIONS

Hybrid AI for Supply Chain Optimization | Presented in Comsys 2026 *2026*

- Co-authored peer-reviewed work combining LSTM/GRU demand forecasting, NSGA-II multi-objective optimisation, and PPO reinforcement learning; presented at BITS 2026

Language Model Behaviour in Natural Language Tasks *2025*

- Co-authored a study on how Large and Small Language Models interpret and fall short of human-level nuance; placed 2nd in two institutional research paper competitions

ATHLETICS, CERTIFICATIONS & LEADERSHIP

Athletics: National Gold Medallist — Archery · National Level — Rifle Shooting · School Sports Captain (2020–21)

Certifications: Qiskit Global Summer School 2026 (Quantum Excellence) · Gemini Multimodal RAG · Prompt Design in Vertex AI · Polarizer (2025) :2nd Place Paper Presentation · Techskript (2025): 2nd Place Paper Presentation

Leadership: IET, MPSTME — Core Member (2025–2026) · Findrome, MPSTME — Executive (2024–25)